

Muhammad Arsalan

Data Extraction Specialist | Python · Scrapy · Selenium

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SUMMARY

Python Developer and Data Extraction Specialist with 3+ years of experience building scalable web scraping and automation solutions using Scrapy, Selenium, BeautifulSoup, and REST APIs. Experienced in extracting, processing, and validating large-scale structured datasets from dynamic and JavaScript-rendered websites. Proven ability to automate complex data collection workflows and deliver reliable high-volume extraction systems for analytics and business operations.

SKILLS

Web Scraping & Data Extraction: Scrapy, Selenium, BeautifulSoup, Requests, Web Scraping, Data Extraction, Dynamic Website Scraping, CAPTCHA Handling, Pagination Handling, Data Cleaning, Structured Data Processing

Languages & frameworks: Python, JavaScript, SQL, Flask, FastAPI, HTML, CSS, Bootstrap

Databases: PostgreSQL, MySQL, MongoDB (NoSQL), Qdrant (Vector DB), Redis

Cloud & Infrastructure: AWS (EC2, RDS, S3), Docker, Linux, CI/CD

Data Processing & Automation: Pandas, NumPy, ETL Pipelines, Workflow Automation, REST APIs, JSON, CSV, Data Validation

Data Analysis & Visualization: Power BI (DAX, OLAP), Streamlit, Matplotlib, Seaborn, Pandas, NumPy, Statistics

AI & LLMs: LangChain, RAG Pipelines, LLM Applications, Prompt Engineering, OpenAI API, Gemini, Hugging Face, Computer Vision

Tools & workflow: Git, Jira, ClickUp, Excel, Google Sheets

PROFESSIONAL EXPERIENCE

Data Engineer

May 2025 – present

OfficeField

- Engineered Apache Airflow-orchestrated pipelines automating real-time stock and crypto data extraction from 5 financial APIs (FMP, Nasdaq, CoinMarketCap, CoinGecko ,etc), reducing data refresh cycles from daily to sub-hourly.
- Automated real-time ingestion and multi-destination storage across MongoDB, PostgreSQL, AWS RDS, and Qdrant Vector DB — processing thousands of financial records daily with zero manual intervention.
- Implemented automated data quality monitoring and validation checks across 4 production storage systems, achieving 99%+ data accuracy for downstream analytics consumers.
- Optimized workflow orchestration and Airflow task scheduling through parallelized ETL execution, improving pipeline performance and SLA compliance across real-time financial data pipelines.

Data Extraction Specialist (Freelancer)

Jul 2024 – Apr 2025

UpWork & Fiverr

- Delivered 30+ web scraping projects using Scrapy, Selenium, and BeautifulSoup — extracting and processing millions of records across paginated, dynamic, and CAPTCHA-protected websites.
- Automated end-to-end data collection workflows, reducing client manual effort by ~80% and delivering structured datasets within agreed SLAs on every project.
- Engineered robust web scraping solutions for JavaScript-rendered and dynamically paginated websites, maintaining 98%+ extraction completeness across high-volume datasets.

Data Engineer

Feb 2023 – Jun 2024

Xloop Digital Services (Pvt) Ltd

- Architected scalable distributed data processing pipelines using PySpark and AWS Glue to transform high-volume customer datasets powering real-time recommendation systems.
- Deployed a real-time Kafka streaming pipeline on AWS EC2 handling high-throughput event streams, integrated with AWS Glue for automated schema cataloging — reducing data availability lag from hours to under 5 minutes.
- Built a Python-based multi-database ETL pipeline to extract, clean, and segment customer data by region — loading USA records into MySQL and UK/India records into PostgreSQL, with structured JSON metadata for downstream processing.
- Designed DAX-optimized Power BI dashboards consolidating multi-source financial complaint datasets, enabling self-service analytics for 5+ stakeholders and eliminating recurring manual reporting.

PROJECTS

GovFinder Web Scraping

(Client Project — Private Repository)

Stack: Python, Scrapy, BeautifulSoup, PostgreSQL, CSV

- Developed an automated scraping tool to extract names, positions, and emails of American government officials from multiple official websites, centralizing data into a searchable structured database.
- Reduced government contact discovery time from hours of manual browsing to seconds of database querying.

Tamimi Markets Product Scraper

github.com/Muhammad-Arsalan-Ai/Tamimi-Markets-Product-Scraper

Stack: Python, Requests, CSV, REST API, JSON

- Engineered a scalable Python-based data extraction pipeline leveraging REST APIs to collect and normalize product data across 26 retail categories, enabling downstream analytics and reporting automation.
- Developed resilient extraction workflows with automated pagination, retry handling, and fault tolerance, improving dataset completeness and reducing scraping failures.

EDUCATION

Bachelor Of Computer Science

2019 – 2023

Sir Syed University of Engineering and Technology

CERTIFICATES

- Data Engineering Bootcamp — Emeritus · 2023
- Generative AI with Large Language Models — Coursera / DeepLearning.AI · 2024
- Certified Artificial Intelligence Developer — PIAIC · 2023
- Introduction to Data Analytics — LinkedIn Learning · 2024